


CHALMERS
UNIVERSITY OF TECHNOLOGY

# Multi-Agentic Systems in HRI (MAgicS-HRI): Bridging Design and Real-World Challenges for End Users

Downloaded from: https://research.chalmers.se, 2026-11-07 01:38 UTC

Citation for the original published paper (version of record):
Zhang, Y., Schömbs, S., Pan, X. et al (2026). Multi-Agentic Systems in HRI (MAgicS-HRI): Bridging Design and Real-World Challenges for End
Users. Companion Proceedings of the 21st ACM IEEE International Conference on Human Robot Interaction Hri Companion 2026: 1369-1371. http://dx.doi.org/10.1145/3776734.3788825

N.B. When citing this work, cite the original published paper.

© 2026 IEEE. Personal use of this material is permitted. Permission from IEEE must be obtained for all other uses, in any current or future media, including reprinting/republishing this material for advertising or promotional purposes, or reuse of any copyrighted component of this work in other works.

(article starts on next page)

# Multi-Agentic Systems in HRI (MAgicS-HRI): Bridging Design and Real-World Challenges for End Users

Yan Zhang
University of Melbourne
Melbourne, VIC, Australia
yan.zhang.1@unimelb.edu.au

Sarah Schömbs
University of Melbourne
Melbourne, VIC, Australia
sschombs@student.unimelb.edu.au

Xiang Pan
Kyoto University
Kyoto, Japan
pan@robot.soc.i.kyoto-u.ac.jp

Jan Leusmann
LMU Munich
Munich, Germany
jan.leusmann@ifi.lmu.de

Sara Mongile
Italian Institute of Technology
Genoa, Italy
sara.mongile@iit.it

Mohammad Obaid
Chalmers University of Technology
and University of Gothenburg
Gothenburg, Sweden
mobaid@chalmers.se

Wafa Johal
University of Melbourne
Melbourne, VIC, Australia
wafa.johal@unimelb.edu.au

## Abstract

Recent developments in large language models and multi-agentic systems present both opportunities and challenges for Human-Robot Interaction (HRI). However, it remains unclear how the interaction paradigms will be reshaped due to the growing complexity and opaqueness. Therefore, to bridge this gap and connect users with multi-agentic systems through embodied robot agents, this workshop aims to explore solutions for design challenges of multi-agentic systems on various types of robots in real-world contexts. Through a complementary mix of focus groups, including design activities and paper presentations, our workshop generates design considerations for real-world applications and outlines pioneering research agendas for multi-agentic systems in HRI.

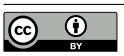
## CCS Concepts

• **Human-centered computing → Interaction design theory, concepts and paradigms**.

## Keywords

Multi-Agentic System, Human-Robot Interaction, Design, Embodiment

**ACM Reference Format:**
Yan Zhang, Sarah Schömbs, Xiang Pan, Jan Leusmann, Sara Mongile, Mohammad Obaid, and Wafa Johal. 2026. Multi-Agentic Systems in HRI (MAgicS-HRI): Bridging Design and Real-World Challenges for End Users. In *Companion Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction (HRI Companion '26), March 16–19, 2026, Edinburgh, Scotland, UK*. ACM, New York, NY, USA, 3 pages. https://doi.org/10.1145/3776734.3788825



This work is licensed under a Creative Commons Attribution 4.0 International License.
*HRI Companion '26, Edinburgh, Scotland, UK*
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2321-6/2026/03
https://doi.org/10.1145/3776734.3788825

## 1 Introduction

With the advances of large language models (LLMs), agentic systems are gaining increasing attention due to their goal-driven directness, long-horizon planning, and general-purpose roles [23]. Companies such as Microsoft [22] and OpenAI [1] have introduced their multi-agentic framework, signally a shift toward agent-oriented programming [19]. However, the rapid growth in system complexity and opaqueness has exposed critical design challenges [17]. New interaction paradigms and design solutions are urgently needed to address these emerging issues.

"Multi-agent" and "Swarm" systems have long been established fields in human-robot interaction (HRI) and robotics [9, 20]. Researchers explored various architectures, including hierarchical and decentralised approaches [1, 11, 22]. However, emerging multi-agentic systems powered by LLMs offer new possibilities for integration with embodied robots. Previous work explored re-embodiment and co-embodiment of agents on a single robot [8]. Robots may also be interconnected through distributed agents from the same underlying multi-agentic system [2]. The distinctive characteristics of agentic systems, including underspecification, directness of impact, goal-directness, and long-term planning [4], demonstrate strong potential to enhance robot intelligence and support higher degrees of agency, enabling robots to engage more effectively in human society.

While the integration of multi-agentic systems with robots is presenting substantial opportunities, it also necessitates careful reconsideration of key embodied properties such as autonomy [3, 25], social ability [16], and anthropomorphism [15, 26]. Importantly, multi-agentic systems in HRI introduce challenges [17], ranging from transparency, parallel interaction, cascading failures, conflicts, cognitive load, and mental models, all of which are essential for novel end-users to effectively interact with robots as embodied interfaces to complex multi-agentic systems. Further, such systems may require new ways of communicating risk (e.g., conflicts between agents in multi-agentic systems [17]) that take their embodied form

into account [14, 16], suddenly presenting robots as conduits for risk communication and as risk communicators.

Therefore, this workshop focuses on addressing these challenges in real-world contexts by bringing together researchers to participate in focus groups and discussions, defining research agendas that will enable multi-agentic robots to benefit end-users and contribute meaningfully to human society.

## 2 Workshop Overview

This workshop aims to pioneer solutions for the design challenges of multi-agentic systems in HRI by integrating insights from both design and technical perspectives. We invite researchers to envision and address potential user tensions through speculative narratives that reveal the consequences of poor design, then collaboratively iterate toward more effective solutions.

### 2.1 Format and Activities

This is a half-day workshop for in-person participants. Participants are encouraged to submit a position paper to share their recent work on the following topics of interest:

- Orchestration interfaces for human oversight of agents and robots
- Failure and trust in human-robot interaction
- Risk communication for inter-robots or human-robot conflicts
- Humans' mental model and cognitive load in HRI with multi-agentic systems
- Anthropomorphism in multi-agentic robots
- Re-embodiment, co-embodiment, distributed embodiment
- Human-multi-agents communication
- Human-robot interaction in real-world scenarios
- LLMs and agents powered human-robot interaction
- Theoretical frameworks for human and multi-agentic interaction

Authors with accepted papers will present their work during the paper madness session using a single slide to provide diverse and multiple perspectives on multi-agentic systems in HRI.

The focus group is structured around card decks inspired by [21], which provides a constructive and narrative approach to the discussion topics. There will be three decks of cards. The **robot deck** presents different types of robots, ranging from conversational agents such as Furhat, social robots such as Pepper, humanoid collaborative robots like TIAGo, to industrial robots such as Franka. The **agent deck** includes multi-agentic architectures, such as multi-body, hierarchical, and decentralised systems, as well as predefined agent roles. The **scenario deck** provides detailed contexts for real-world human–robot interaction scenarios, including domestic, industrial, and healthcare settings. The **challenge deck** introduces design challenges for multi-agentic systems derived from [17].

During the focus group, participants will be divided into small groups. In *Phase 1*, each group will draw a card from the scenario deck and use cards from the robot and agent decks as inspiration to design a multi-agentic robots for that scenario. In *Phase 2*, pairs of groups will exchange their design ideas. Each group will then use cards from the challenge deck to articulate end-user challenges and "attack" the other group's design. In *Phase 3*, the pairs of groups will exchange their challenges and iterate on their designs to "defend" them by developing solutions. After the focus group, participants will engage in a collective reflection session to discuss the activity, refining the design solutions developed under different scenarios into general design principles for multi-agentic systems in HRI.

### 2.2 Structure

**13:30 - 13:50** Welcome, introduction, and ice-breaking
**13:50 - 14:20** Paper madness
**14:20 - 14:35** Focus group activity introduction
**14:35 - 15:00** Focus group phase one
**15:00 - 15:30** Coffee break
**15:30 - 15:55** Focus group phase two
**15:55 - 16:35** Focus group phase three
**16:35 - 16:50** Discussion and reflection
**16:50 - 17:00** Conclusion remarks

## 3 Solicitation Plan, Logistics, and Resources

We aim to bring together a diverse community of individuals with an interest in multi-agentic systems within the field of human–robot interaction. Participation is welcomed from researchers, industry professionals, and stakeholders spanning disciplines such as HRI, design, robotics, psychology, AI, and related interdisciplinary domains that explore both the design and technical aspects of HRI. As the main format of this workshop will be focus groups, the maximum number of participants will be approximately 50, with up to 10 accepted position papers for presentation. Authors of accepted papers will be invited to give a three-minute presentation with an emphasis on fostering collaborative discussion during the 'paper madness' session.

The call for participation will be disseminated through academic mailing lists and social media platforms, including but not limited to HRI-Announcement, CHI-Announcements, Robotics Worldwide, LinkedIn, and X. To broaden engagement, we will also share the call via lists that extend beyond the immediate HRI community, such as HMC (for scholars in human–machine communication), and the Women in Machine Learning network.

The workshop requires a meeting room for 50 participants and equipped with audio-visual facilities such as a projector, microphone, and speakers. The organising committee will be responsible for preparing the card deck and stationery for the focus group sessions.

## 4 Documentation

A dedicated workshop website [1] will be developed to provide detailed information, including the program format, schedule, and organising committee. Participants will be encouraged to submit a four-page position paper with additional space permitted for references on topics of interest. All papers will undergo peer review based on originality, relevance, and clarity. Submissions offering bold and critical insights are especially encouraged.

The titles and abstracts of accepted papers will be displayed on the workshop website prior to the conference. The workshop's event logs and photos will be documented and made available on the website.

## 5 Organising Committee

*Yan Zhang*. (https://yzhang2332.github.io) is a 3rd year PhD student at the University of Melbourne. Her current research focuses on the natural communication in human-robot physical collaboration and how multi-agentic systems enhance interaction [17, 24, 26]. She is part of the organising committee for HRI'26 and has co-organised HRI workshops.

[1] https://sites.google.com/view/magics-hri/home

*Sarah Schömbs.* (https://sites.google.com/view/sarahschombs) is a final year PhD student at the University of Melbourne researching human-centred risk communication in Human-Agent Interaction. She examines how users perceive risks communicated through agents [14, 18], including system uncertainties [16], with a focus on AI risks and agentic systems [17]. She was part of the organising committee for HRI'25 and has co-organised HRI workshops.

*Jan Leusmann.* (https://leusmann.io) is a 3rd year PhD student at LMU Munich. His research is situated in the domain of HRI and focuses on fostering understandable and intuitive communication between robots and humans, e.g., [5–7]. Recently, he focused on creating and understanding curious robot behaviour.

*Xiang Pan.* (https://pan-xiang.github.io) is a post-doctoral researcher at Kyoto University. His research focuses on human-robot interaction and embodied AI, with an emphasis on designing expressive manipulation to improve the transparency and naturalness of robot actions in human-shared environments [12, 13].

*Sara Mongile.* (https://iit.it/it/people/sara-mongile) is a postdoctoral researcher at the Italian Institute of Technology. In her research, she investigates how to develop cognitive architectures for social robots to promote better human-robot interactions and foster robots' autonomous and adaptive behaviour [10].

*Wafa Johal.* (https://wafa.johal.org) is an associate professor at the University of Melbourne. Her research combines concepts and methods from AI and HCI to create acceptable and useful robots to learn, teach and collaborate using multimodal interaction. She was the General Chair of HRI'25 and has co-organised various conferences and workshops.

*Mohammad Obaid.* (https://mohammadobaid.se) is an Associate Professor of HCI in the Department of Computer Science and Engineering at Chalmers University of Technology, Sweden. He has co-authored more than 100 papers on HCI and HRI across a range of topics. He has served on organizing and program committees for leading HCI and HRI conferences like CHI, HRI, HAI, and NordiCHI.

## Acknowledgments

We acknowledge support from the Australian Research Council for projects CE260100108, DP260101082, and FT250100459.

## References

[1] 2025. openai-agent. https://github.com/openai/openai-agents-python/tree/main

[2] Karla Bransky, Penny Sweetser, Sabrina Caldwell, and Kingsley Fletcher. 2024. Mind-body-identity: A scoping review of multi-embodiment. In *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction*. 65–75.

[3] Cristiano Castelfranchi. 1994. Guarantees for autonomy in cognitive agent architecture. In *International Workshop on Agent Theories, Architectures, and Languages*. Springer, 56–70.

[4] Alan Chan, Rebecca Salganik, Alva Markelius, Chris Pang, Nitarshan Rajkumar, Dmitrii Krasheninnikov, Lauro Langosco, Zhonghao He, Yawen Duan, Micah Carroll, Michelle Lin, Alex Mayhew, Katherine Collins, Maryam Molamohammadi, John Burden, Wanru Zhao, Shalaleh Rismani, Konstantinos Voudouris, Umang Bhatt, Adrian Weller, David Krueger, and Tegan Maharaj. 2023. Harms from Increasingly Agentic Algorithmic Systems. In *2023 ACM Conference on Fairness, Accountability, and Transparency*. ACM, Chicago IL USA, 651–666.

[5] Jan Leusmann, Anna Belardinelli, Luke Haliburton, Stephan Hasler, Albrecht Schmidt, Sven Mayer, Michael Gienger, and Chao Wang. 2025. Investigating LLM-Driven Curiosity in Human-Robot Interaction. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 599, 16 pages.

[6] Jan Leusmann, Ludwig Felder, Chao Wang, and Sven Mayer. 2025. Understanding Preferred Robot Reaction Times for Human-Robot Handovers Supported by a Deep Learning System. In *Proceedings of the 2025 ACM/IEEE International Conference on Human-Robot Interaction (HRI '25)*. IEEE Press, 4–13.

[7] Jan Leusmann, Steeven Villa, Thomas Liang, Chao Wang, Albrecht Schmidt, and Sven Mayer. 2025. An Approach to Elicit Human-Understandable Robot Expressions to Support Human-Robot Interaction. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 534, 17 pages. doi:10.1145/3706598.3713085

[8] Michal Luria, Samantha Reig, Xiang Zhi Tan, Aaron Steinfeld, Jodi Forlizzi, and John Zimmerman. 2019. Re-Embodiment and Co-Embodiment: Exploration of social presence for robots and conversational agents. In *Proceedings of the 2019 on Designing Interactive Systems Conference*. 633–644.

[9] Maja J. Matarić. 1995. Issues and approaches in the design of collective autonomous agents. *Robotics and Autonomous Systems* 16, 2-4 (Dec. 1995), 321–331. doi:10.1016/0921-8890(95)00053-4

[10] Sara Mongile. 2025. Towards Adaptive Human-Robot Interaction: A Comfort-Driven Framework for Social Robots. In *2025 20th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. IEEE, 1872–1874.

[11] Jun Ota. 2006. Multi-agent robot systems as distributed autonomous systems. *Advanced engineering informatics* 20, 1 (2006), 59–70.

[12] Xiang Pan, Malcolm Doering, and Takayuki Kanda. 2024. What is your other hand doing, robot? A model of behavior for shopkeeper robot's idle hand. In *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction*. 552–560.

[13] Xiang Pan, Malcolm Doering, Stela Hanbyeol Seo, and Takayuki Kanda. 2025. Communicating Physical Properties Through Robot Object Manipulation. In *2025 20th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. IEEE, 738–746.

[14] Sarah Schömbs, Jorge Goncalves, and Wafa Johal. 2025. "I can feel the risks by looking at the robot face": Communicating Risk through a Physical Agent. In *Proceedings of the 2025 ACM Designing Interactive Systems Conference (DIS '25)*. Association for Computing Machinery, 236–252. doi:10.1145/3715336.3735759

[15] Sarah Schömbs, Jiahe Pan, Yan Zhang, Jorge Goncalves, and Wafa Johal. 2024. FaceVis: Exploring a Robot's Face for Affective Visualisation Design. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*. Association for Computing Machinery, Article 161, 10 pages.

[16] Sarah Schömbs, Saumya Pareek, Jorge Goncalves, and Wafa Johal. 2024. Robot-Assisted Decision-Making: Unveiling the Role of Uncertainty Visualisation and Embodiment. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 589, 16 pages. doi:10.1145/3613904.3642911

[17] Sarah Schömbs, Yan Zhang, Jorge Goncalves, and Wafa Johal. 2025. From Conversation to Orchestration: HCI Challenges and Opportunities in Interactive Multi-Agentic Systems. In *Proceedings of the 13th International Conference on Human-Agent Interaction (HAI '25)*. Association for Computing Machinery, Yokohama, Japan. doi:10.1145/3765766.3765795

[18] Sarah Schömbs. 2025. Communicating Internal and External Risks in Human-Robot Interaction. In *2025 20th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. 1881–1884. doi:10.1109/HRI61500.2025.10973977

[19] Yoav Shoham. 1993. Agent-oriented programming. *Artificial intelligence* 60, 1 (1993), 51–92.

[20] Cathérine Tessier, Laurent Chaudron, and Heinz-Jürgen Müller. 2005. *Conflicting Agents: Conflict Management in Multi-Agent Systems.* Springer Science & Business Media. Google-Books-ID: YODlBwAAQBAJ.

[21] Luis Von Ahn and Laura Dabbish. 2008. Designing games with a purpose. *Commun. ACM* 51, 8 (2008), 58–67.

[22] Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Shaokun Zhang, Erkang Zhu, Beibin Li, Li Jiang, Xiaoyun Zhang, and Chi Wang. 2023. Autogen: Enabling next-gen llm applications via multi-agent conversation framework. *arXiv preprint arXiv:2308.08155* (2023).

[23] Zhiheng Xi, Wenxiang Chen, Xin Guo, Wei He, Yiwen Ding, Boyang Hong, Ming Zhang, Junzhe Wang, Senjie Jin, Enyu Zhou, et al. 2025. The rise and potential of large language model based agents: A survey. *Science China Information Sciences* 68, 2 (2025), 121101.

[24] Yan Zhang. 2025. Implicit Communication of Contextual Information in Human-Robot Collaboration. In *2025 20th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. 1903–1905. doi:10.1109/HRI61500.2025.10974036

[25] Yan Zhang, Ziang Li, Haole Guo, Luyao Wang, Qihe Chen, Wenjie Jiang, Mingming Fan, Guyue Zhou, and Jiangtao Gong. 2023. "I am the follower, also the boss": Exploring Different Levels of Autonomy and Machine Forms of Guiding Robots for the Visually Impaired. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, Article 542, 22 pages. doi:10.1145/3544548.3580884

[26] Yan Zhang, Tharaka Sachintha Ratnayake, Cherie Sew, Jarrod Knibbe, Jorge Goncalves, and Wafa Johal. 2025. Can you pass that tool?: Implications of Indirect Speech in Physical Human-Robot Collaboration. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, Article 781, 18 pages. doi:10.1145/3706598.3713780

Received 2025-10-10; accepted 2025-11-21